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SUPERSEDING
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MILITARY SPECIFICATION

DEVICES, METAL: BREAST AND COLLAR, GENERAL SPECIFICATION FOR

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements of metal breast and collar insignia used by the Department of the Navy.

* 1.2 Classification. The devices covered by this specification shall be as specified on the applicable Military Specification Sheets or current supplement. (see 2.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8455

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SPECIFICATIONS

FEDERAL

NN-P-71 - Pallet, Material Handling, Wood, Stringer Construction 2 Way and 4 Way Entry
PPP-B-566 - Box, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-665 - Box, Paperboard, Metal Edges and Components
PPP-B-676 - Boxes, Setup

MILITARY

MIL-F-495 - Finish, Chemical, Black, for Copper Alloys
MIL-P-15011 - Pallet, Material Handling, Wood, Post Construction, 4 Way Entry

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

SPECIFICATION SHEETS

MIL-D-16475/ - (For applicable military specification sheet see current supplement)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

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* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

THE INSTITUTE OF HERALDRY, US ARMY STANDARD COLOR CHIP SET

(Referenced Standard Color Chips may be obtained from the procuring activity.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B-487 - Measurement of Metal and Oxide Coating Thickness

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail requirements. Detail requirements or exceptions applicable to a particular insignia shall be as specified on the applicable specification sheet.

3.2 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.2). Variations from this specification may appear in the sample in which case this specification shall govern.

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3.2.1 Government loaned property. Hubs, whe furnished by the Government, shall be used only for making the contractor's working dies. The contractor's stamping and embossing dies shall be tooled and polished to remove dents, scratches, and all other imperfections (see 6.3).

3.2.2 Approved contractor sample. When hubs are not available, all requirements specified on the applicable specification sheet and as exhibited by the approved contractor's sample shall apply (see 6.4).

3.3 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.4 Material.

3.4.1 Copper base alloys. The chemical composition of the copper base alloys shall conform to the requirements of Table I, when tested as specified in 4.4.1. The copper base alloys shall be roll polished, free from pits, scale (including red oxide), dents, nicks, scratches, segregations and foreign inclusions that will not be removed in later processes (see 4.4.1).

Table I - Chemical composition of copper base alloys (percent)

Alloy	Copper	Nickel	Lead (max)	Iron (max)	Zinc	Others (max)
Yellow Brass	64.0-68.5	-	0.15	0.05	Remainder	0.15
Red Brass	84.0-86.0	-	0.05	0.05	Remainder	0.15
Nickel Silver	63.0-66.5	16.5-19.5	0.10	0.25	Remainder	0.50
Nickel Silver (pin only)	53.5-56.5	16.5-19.5	0.05	0.25	Remainder	0.50
Gilding Metal	94.0-96.0	-	0.03	0.05	Remainder	0.13

3.4.2 Gold.

3.4.2.1. Gold-plating. Gold for plating shall be 24 Karat (see 4.4.1).

3.4.3 Silver.

3.4.3.1 Sterling silver. When specified, sterling silver insignia shall be not less than 92.5 percent pure silver (see 4.4.1).

3.4.3.2 Silver-filled. When specified, silver-filled insignia shall be 1/20 silver-filled (front only) over a copper-base alloy. The silver shall be not less than 92.5 percent silver (see 4.4.1).

3.4.3.3 Silver plating. When specified, silver for plating shall be not less than 99.5 percent pure silver (see 4.5).

3.4.4 Enamel, hard. When specified, hard enamel shall be a hard glass type vitreous enamel fused with metallic oxides to produce the desired color, shade and opacity or translucence (see 4.4.1).

3.4.5 Hard solder. Hard solder shall be a silver alloy solder having a

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melting point of not less than 1075 degrees Fahrenheit (see 4.4.1).

3.4.6 Lacquer. The lacquer shall be a pale, clear, synthetic lacquer. The use of a pigmented lacquer shall not be permitted (see 4.4.1).

* 3.4.7 Enamel, flat black. When specified, the enamel specified in 3.4.4 shall have a dead flat finish (see 4.4.1).

* 3.4.8 Chemical finish. The black chemical finish shall conform to MIL-F-495.

3.5 Design. The designs of the insignia shall be exact replicas of the design of the Government loaned hubs (see 3.2.1) or available guide samples (see 3.2).

3.6 Construction.

3.6.1 Stamping, trimming, and piercing. The insignia shall be struck in a manner that will insure a well defined die-struck edge. Insignia shall be trimmed, and when applicable, pierced to the die-struck outline. All edges shall be clean, smooth, and free from burrs, drag, steps, and rough edges. The stamping, trimming, and piercing operations shall not damage or distort the shape or design of the insignia.

* 3.6.2 Attaching device. The attaching device shall be prong and clutch type as shown in Figure 1 and as specified herein. The prong shall be made of nickel silver specified in 3.4.1, 5/16 (± 1/64) inch long, tapered on one end with a slight notch in its diameter located 3/32 (± 1/64) inch from the tapered point. The overall diameter of the prongs shall be 0.044 (± .002) inch conforming to B. A. Ballou No. 34, or equal (see 6.5 and 6.6). The prongs shall be hard soldered or electronically fused to the back of the insignia as specified on the applicable specification sheet. Clutches shall be of the wing type and may be made from any type brass. The overall diameter shall be 15/32 (± 1/32) inch conforming to B. A. Ballou No. 56, or equal (see 6.5 and 6.6). When the size of the insignia is a limiting factor, the clutches shall be 3/8 (± 1/32) inch in diameter. When tested as specified in 4.5, the clutch shall not be removable by hand except when the release wings are depressed.

3.6.3 Soldering or electronic fusion. Soldered or fused joints shall be strong, clean, smooth, and free of burned or reduced areas. Soldered joints shall be free of excess solder and flux. All joints shall be either hard soldered or electronically fused. When solder is used, the material specified in 3.4.5 shall be used. When tested as specified in 4.5, the soldered prong shall not separate from the insignia. Fused joints shall not separate when tested as specified in 4.5.

3.6.4 Backs of insignia. Unless otherwise specified on the applicable specification sheet, the backs of all insignia shall be flat. When a forcer is used to bring up the design, the impression of the forcer on the back of the insignia shall be regular, symmetrical and leave no sharp corners or edges.

3.7 Finish. Unless otherwise specified on the applicable specification sheet, the front and edges (surfaces visible from the front) of all insignia shall be finished to match the guide sample (see 3.2).

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* 3.7.1 Gold plating. Gold specified in 3.4.2.1 shall be used for gold plating. Prior to plating, the insignia shall be cleaned of matter that may affect the gold plating. The plating shall be non-porous and continuous over the entire surface and there shall be no cut through, peeled or blistered plating. The back of the insignia shall be bright and clean. The base metal shall be either yellow brass or red brass specified in 3.4.1. No visible chemical reaction shall appear when tested as specified in 4.5.

3.7.2 Silver plating. Prior to plating, the insignia shall be clean of all matter that may affect the silver plating. The silver used for silver plating shall be as specified in 3.4.3.3. Brass items shall be hot nickel underplated a minimum of 0.00025 inch thick. Silver filled items shall not be underplated. The silver plating shall be a minimum 0.0003 inch thick and shall be smooth, fine grained, adherent, and free from visible defects (see 4.5).

3.7.3 Rhodium plating. When rhodium plating is specified, any copper base alloy may be used which has an underplating of nickel a minimum of 0.00025 inch thick (see 4.4.1). The nickel plated device shall be burnished before plating. When tested as specified in 4.5, no visible chemical reaction shall appear (see 4.5). Silver filled material shall be nickel plated prior to rhodium plating.

3.7.4 Lacquering. Insignia shall be lacquered as specified on the applicable specification sheet. The lacquer film shall be continuous, level, adherent, and free from dust, lint and other foreign inclusions. When tested as specified in 4.5, there shall be no change in the lacquered face of the insignia except for a slight yellowing of the highlights. When tested as specified in 4.5, the tissue paper shall not adhere to the lacquered surface.

3.7.5 Enameling, hard. When specified, the enamel shall be vitreous, charged, fired, stoned level with the dikes and lapped, felted, or wheel polished to produce a hard glass-like finish. The enameling shall be adherent, free from bubbles, pits, cracking, crazing, burned edges, foreign inclusions or other defects that may affect appearance. Enamel colors shall conform to the shades designated by the color chip numbers specified in the Institute of Heraldry, US Army Standard Color Chip Set (see 2.2) and shall match the enamel on the guide sample (see 3.2).

* 3.7.6 Enameling, flat black. When specified, the face and outer edges of the chemically finished insignia shall be thoroughly coated with the enamel specified in 3.4.7. The baked enamel shall not flake or chip when tested as specified in 4.6.8.

* 3.7.7 Chemical finishing. When specified, all surfaces and edges of insignia, before enameling, shall be treated with the black chemical finish specified in 3.4.8.

3.8 Dimensions. All dimensions and tolerances shall be as specified on the applicable specification sheet. When a Government loaned hub is available for use, the hub shall control.

3.9 Marking for identification. The contractor shall stamp his trademark or other identifying mark, legibly and inconspicuously on the back of each

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insignia. In addition, precious metal insignia shall bear the fineness of quality marking.

3.9.1 Cleaning of insignia. Each card used for mounting insignia (see 5.1.1) shall have the following statement printed on the face of the card:

TO CLEAN, RUB GENTLY WITH A SOFT DRY CLOTH.
DO NOT CLEAN WITH A CHEMICALLY TREATED CLOTH OR ABRASIVE.

3.10 Workmanship. The finished insignia shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article submitted in accordance with 3.3, shall be inspected as specified in 4.4.3 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the

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requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition, testing shall be performed on materials as listed in Table II. All test reports shall contain the individual values utilized in expressing the final results. Requirements are applicable to the lot average. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

Table II - Testing of components

Component and unit of product	Characteristic	Requirement paragraph	Test Method	Sample Unit
Copper base alloy (as applicable 1 lb.)	Chemical composition	3.4.1	FED-STD 151 Method 111 or 112	4 oz. (composite)

Determinations per sample unit	Results reported as
2 for each element	Nearest 0.1% each element

In addition to Table II testing, the contractor shall submit a certificate of compliance for the materials and characteristics listed below.

Table III - Instructions for component testing

Component	Characteristic	Requirement paragraph
Gold (for plating)	Karat and base metal	3.4.2.1 and 3.7.1
Sterling Silver	Fineness	3.4.3.1
Silver-filled material 1/	Fineness and weight of lamination	3.4.3.2
Silver (for plating)	Fineness	3.4.3.3
Enamel (hard)	Material Identification	3.4.4
Hard solder	Material Identification	3.4.5
Lacquer	Material Identification	3.4.6
Enamel (flat black)	Finish	3.4.7
Attaching device	Material Identification	3.6.2
Rhodium (for plating)	Material Identification and underplating	3.7.3

1/ In the event of dispute, the material shall be tested for silver content in accordance with 4.6.7.

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4.4.2 In-process inspection. Inspection may be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are carried out as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.3 Examination of the end item. Examination of the end item shall be in accordance with 4.4.3.1 through 4.4.3.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.3.3. The sample unit shall be one finished insignia. The lot shall be expressed in terms of the sample unit.

4.4.3.1 Visual examination. The finished insignia shall be examined for defects in design, workmanship, color, finish, and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

Examine	Defect	Classification Major	*	Minor
I. FRONT				
a. Color or Finish	1. Does not compare favorably with approved contractor sample or first article		*	
	2. Oxidized finish has not been sufficiently relieved or has been excessively relieved (when required)		*	
	3. Foreign matter imbedded		*	
	4. Background not finished as specified	X		
b. Lacquer	1. Missing, not smooth, flakey, blistered peeling, tacky, cloudy, foreign matter embedded			X
	2. Not set to touch			X
c. Enamel	1. Missing where required	X		
	2. Cracked, chipped or crazed		*	
	3. Bubbles, lumps, or foreign matter		*	
	4. Not level with dikes			X
	5. Incorrect color	X		
	6. Not polished as specified (when required)			X
d. Quality of metal	Surface pitted, porous, nicked, scaled, scratched, and dented	X		
e. Design	1. Construction or design details altered or obscured or not conforming to Government hub or approved contractor sample	X		
	2. Warped, twisted or distorted	X		

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Examine	Defect	Classification Major	*	Minor
	3. Detail struck over, resulting in a double impression		*	
II. BACK				
a. Plating	1. Foreign matter imbedded		*	
	2. Not plated when required	X		
	3. Plating not smooth, porous, continuous			X
	4. Peeling, flakey, chipped			X
	5. Not specified type	X		
b. Lacquer	Lacquer missing, tacky, not adherent, foreign matter imbedded			X
c. Fastening device	1. Not specified type	X		
	2. Any component part missing	X		
	3. Not positioned as specified		*	
	4. Pin bent, affecting use		*	
	5. Clutch fasteners do not function properly		*	
	6. Not attached in specified matter	X		
d. Marking	1. Missing, incorrect, illegible, misspelled, not stamped, not placed inconspicuously			X
	2. Fineness marking missing (when applicable)	X		
III. FRONT, BACK AND EDGES				
a. Workmanship and Construction	1. Surface or edge not clean, not smooth, or free from burrs, roughness, drag, step or tool marks		*	
	2. Metal marks on exposed surface such as nick, dent, dig, gouge, or scratch		*	
	3. Edge not die-struck, when applicable	X		
	4. Not trimmed to the die-struck edge, when applicable		*	
	5. Die-struck edge, rim or outline is not well defined, when applicable	X		
	6. Pierced edge not clean and smooth, e.g., any burr, drag, cutter, or file mark, when applicable		*	
	7. Design or shape is damaged, distorted, or altered by piercing operation, when applicable	X		
	8. Forcer used when not specified or insignia forced in other than specified location		*	

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Examine	Defect	Classification Major	*	Minor
	9. Impression of the forcer (when applicable) is not symmetrical with the design on the front			X
	10. Metal torn or broken out by use of forcer			X
	11. Any component missing, bent, twisted, broken, or malformed		*	
	12. Solder does not completely unite parts, i.e., any perceptible opening between component parts		*	
	13. Area burned or reduced in soldering		*	
	14. Soldered joint not clean or smooth			X
	15. Excess solder or excess flux not removed			X

4.4.3.2 Dimensional examination. Finished insignia shall be examined for defects in dimension. Any dimension not within the tolerance specified on the applicable specification sheet shall be classified as a defect.

4.4.3.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and acceptable quality levels expressed as defects per hundred units are as follows:

	Inspection level	AQL Major	AQL Total
For visual examination in 4.4.3.1	II	2.5	15.0
For dimensional examination in 4.4.3.2	S-3	One class	4.0

4.4.4 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or

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less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

* 4.4.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

* 4.5 Testing of the end item. Completely fabricated metal insignia shall be tested for the characteristics listed in Table IV. The sample unit for the test for fused joints shall be one and the inspection level shall be S-2 and the AQL 6.5. The sample unit for all other applicable end items shall be three (3), the inspection level shall be S-1 and the AQL 6.5. Requirements shall be applicable to the individual unit. Regardless of the number of prong failures on sample unit, it shall be classified as one defective unit.

Table IV - Testing of the end item

Characteristic	Requirement paragraph	Test method
Test for attaching device	3.6.2	4.6.4
Test for soldered prongs or joints	3.6.3	4.6.5
Test for fused prongs or joints	3.6.3	4.6.6
Test for gold plating	3.7.1	4.6.3
Test for nickel underplating thickness	3.7.2	Std com'l 1/
Test for silver plating thickness	3.7.2	Std com'l 1/
Test for rhodium plating	3.7.3	4.6.3
Test for lacquer 2/		
Liver of sulfur	3.7.4	4.6.2
Tackiness	3.7.4	4.6.1
Test for silver content	3.7.2	4.6.7
Test for enamel	3.7.6	4.6.8

1/ In case of dispute, thickness shall be determined in accordance with ASTM Method B 487.

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2/ A certificate of compliance shall be submitted stating that insignia has been lacquered as required.

4.6 Tests.

4.6.1 Tackiness test. A piece of tissue paper shall be pressed against the finished lacquered metal device while at room temperature (60° to 80°F) for 15 seconds with any pressure capable of being exerted by the thumb and two fingers. At the completion of 15 seconds, the finger pressure shall be released to determine compliance with 3.7.4.

4.6.2 Liver of sulfur test for lacquer. (Not applicable to gold insignia). The insignia shall be immersed in a 2 percent by weight of CP liver of sulfur (sulfated potash) water solution at a temperature of 100° ± 10°F for a period of 3 minutes. The insignia shall then be removed and rinsed in warm, then cold, then hot water after which it shall be wiped gently with an absorbent cellulose material or whirled to remove residual moisture, allowed to air dry at room temeperature (60° to 80°F) for 1 hour and examined to determine compliance with 3.7.4.

* 4.6.3 Acid test for gold or rhodium plating. Insignia may be unlacquered or have the lacquer completely removed with an organic solvent. The acid test shall be conducted as follows: Using a test acid consisting of a solution containing 50 percent by volume nitric acid (specific gravity 1.42) and an equal volume of distilled water. At room temperature place a drop of test acid (each drop to be 1/8 inch minimum diameter on three different spots on the face of the insignia). The drops shall remain on the insignia for not less than one (1) minute for Navy items and not less than one and one half (1 1/2) minutes for Coast Guard items. A minimum of two of these spots shall withstand this test.

4.6.4 Test for clutch type fastening devices. The clutch fastener shall be removed from each prong of the attaching device and replaced 10 times, after which it shall not be possible to remove any clutch fastener from its prong except by depressing the release wings.

4.6.5 Solder joint test. Samples to be tested shall be placed in an oven maintained at 1075°F (± 5°F) for not less than 15 minutes. While at the specified oven temperature, samples shall be lifted by the attaching device and shall be inspected for compliance with 3.6.3.

4.6.6 Test for electronically fused joints. The item to be tested shall be anchored on a horizontal surface. The prong shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90° (45 degrees to each vertical) until separation occurs. The fused joint shall then be examined to determine compliance with 3.6.3. A bending tool in the form of a 45° template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.6.7 Silver content. Transfer a 0.5 gram sample, accurately weighed and free of rhodium plating, to a 250 ml beaker. Add 20 ml of 1:4 nitric acid (HNO_3) warm until solution is complete, and boil to expel brown fumes. Cool, transfer to a 1000 ml volumetric flask, and dilute to the mark. Pipette a 200 ml aliquot from the flask into a 400 ml beaker. Heat to boiling, and add HCL

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(3:100) drop by drop from a pipette until precipitation is complete (1 ml of HCL solution precipitates about 0.038 grams of silver). When solution has cooled and precipitation has settled, determine whether precipitation is complete by adding a few drops of the HCL (3:100). If solution remains clear, filter on a weighed porcelain gooch crucible. Wash the precipitate several times with HNO_3 (1:200) and finally with cold water. Protect the silver chloride from the light; dry in an oven for not less than two hours, cool and weigh. The silver content shall be calculated as follows:

$$\frac{W\ (AGCL)\ X\ 752.63X5X100}{WM} = P$$

Whereas:

W(AGCL) = Weight of silver chloride
WM = Weight of metal
P = Percent of silver

* 4.6.8 Test for baked enamel. Draw a knife edge across the face side of the specimen, making four parallel scores one sixteenth inch apart. Repeat this process of drawing the knife edge across the surface but this time perpendicular to the previously scored lines, making small squares in the enamel finish. No flaking or chipping of the small squares shall occur.

5. PACKAGING

5.1 Preservation packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

* 5.1.1.1 Unit packaging. One breast insignia or one pair of collar insignia shall be mounted with its own attaching device on a white sulphite cardboard having a minimum thickness of 0.011 inch. The cardboard shall be cut to a size extending approximately 1/2 inch beyond outer edges of insignia. Each mounted insignia shall be individually packaged in a snug fitting clear polyethylene film bag having a minimum thickness of 0.002 inch. All seams and closures shall be heat sealed. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

* 5.1.1.2 Intermediate packaging. One hundred (100) cardboards with insignia attached as specified in 5.1.1.1 shall be placed in a setup paperboard box conforming to Type I, Variety 1, Class A, Style 4 of PPP-B-676; folding paperboard box conforming to Variety 1, Style III, Type G, Class 1, Sub-class 1 of PPP-B-566; or metal stayed box conforming to Class 1, Style A of PPP-B-665.

* 5.1.2 Level C. Insignia shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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* 5.2.1 Level A. Six-hundred metal insignia, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to Type CF, Class Weather-resistant, Variety DW, Grade V15C, Style RSC of PPP-B-636. Each shipping container shall be closed in accordance with Method III, waterproofed in accordance with Method V, and reinforced in conformance with the appendix of the box specification.

* 5.2.2 Level B. Six-hundred metal insignia, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to Type CF, Class Domestic, Variety DW, Grade 200, Style RSC of PPP-B-636. Each shipping container shall be closed in accordance with Method II of the appendix of the box specification.

5.2.3 Level C. Item packaged, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.2.4 Palletization. When specified (see 6.2) item, packed as specified shall be palletized on a 4-way entry pallet in accordance with load type IA of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means C, K, and L or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, type IV; Type V, class 1, size 2; or Type VIII, fabricated from wood groups I, II, III, or IV, Grade A of NN-P-71, or 4-way, style 1, Size A, Type I, Class 1 fabricated from wood groups specified, of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers metal breast and collar insignia worn by officers and enlisted personnel of the Navy, Coast Guard and Coast Guard Auxiliary, as indicated on the applicable specification sheet.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. The number and date of the applicable specification sheet and supplement (see 1.2 and 2.1).
c. Availability of guide sample (see 3.2).
d. When first article sample is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquistion

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documents regarding arrangements for examination, quantity and testing and approval.

e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

f. When palletization is required (see 5.2.4).

6.3 Government loaned hubs. For access to Government loaned hubs, address the procuring activity or Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 (see 3.2.1).

6.4 Approved contractor sample. A contractor submitting a sample for Government approval shall forward the sample to the procuring activity or to Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490.

* 6.5 Prong and clutch fasteners. Prong and clutch type fasteners, as described in 3.6.2 may be obtained from B. A. Ballou & Company Inc., 800 Waterman Ave, East Providence, R. I. 02914.

* 6.6 Equal item. Prior to the use of an "or equal" item, (see 3.6.2) the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval.

6.7 This specification shall be used in conjunction with the applicable supplement and specification sheet or other document which lists the specific requirements for the insignia to be procured.

* 6.8 Subject term (key word) listing.
breast device
clutch
collar device
insigna
metal
prong

6.9 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy - NU

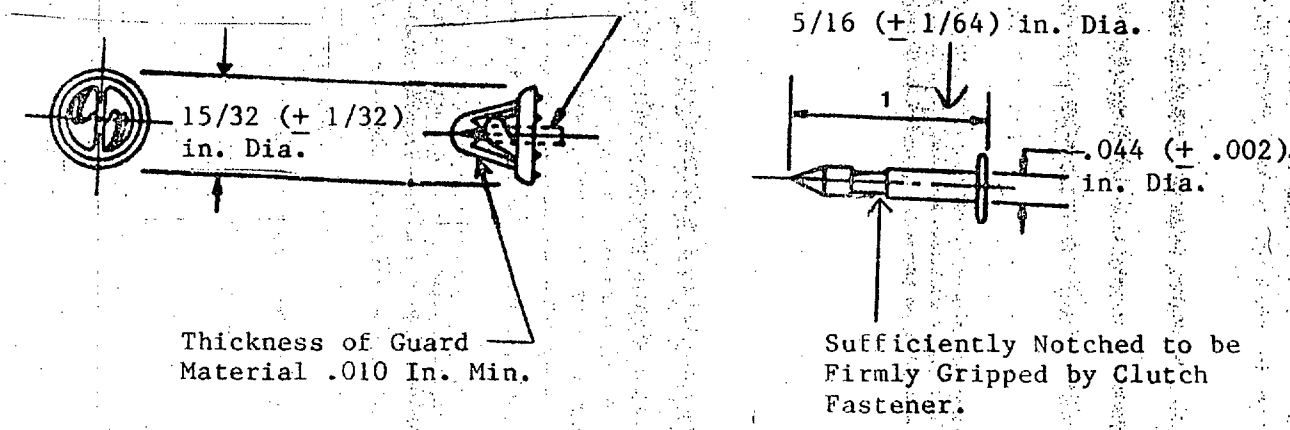
User Activity:
Navy - CG

Review Activity:
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ABOVE MEASUREMENTS ARE IN INCHES

PRONG AND CLUTCH TYPE

Figure 1 - Metal Attaching Devices

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